

Sum and Difference Identities

Date_____ Period____

Find the exact value of each.

1) $\sin -15$

2) $\sin 165$

3) $\cos \frac{17\pi}{12}$

4) $\cos -105$

5) $\tan 105$

6) $\tan \frac{5\pi}{12}$

7)
$$\frac{\tan \frac{8\pi}{9} - \tan \frac{\pi}{18}}{1 + \tan \frac{8\pi}{9} \tan \frac{\pi}{18}}$$

8) $\cos 10 \cos 35 - \sin 10 \sin 35$

9)
$$\frac{\tan 76 + \tan 164}{1 - \tan 76 \tan 164}$$

10)
$$\sin \frac{\pi}{9} \cos \frac{2\pi}{9} + \cos \frac{\pi}{9} \sin \frac{2\pi}{9}$$